

Motivation and Interest

My enthusiasm for experimental quantum sciences began during my projects involving quantum mechanics. My 4th-year project was on quantum weak values which interested me as weak values can amplify signals in metrology experiments. Drawn to this research domain, I have taken dedicated modules for quantum mechanics in my final year. Additionally, my final year project focused on Casimir-Polder forces which interested me due to potential applications in nano-sensing.

Outside my degree, I have done research in multiple domains of experimental physics. One domain is in experimental particle physics focusing on semiconductor sensors called Low Gain Avalanche Detectors (LGAD) for CERN's LHC experiment. In this work, I was invited to join the UK's LGAD development group consisting of researchers from partner institutions of Glasgow, Edinburgh, Manchester, AGH Kraków, CERN, and IMB-CNM. I found my involvement in experimental physics research especially fulfilling due to its challenge requiring adaptability and tenacity to push it forward. Therefore, by taking on a PhD project at Durham University, I want to challenge myself to the limit and improve my research skills.

My intention is therefore to combine my passion for quantum sciences and experimental physics and pursue a PhD project in quantum metrology. One example of research I am particularly interested in is the optical clock array for application in quantum metrology. Recent research has demonstrated an optical lattice clock (OLC) based on strontium having the highest observed precision in the order of 10^{-18} [1]. In parallel, it is known that entangled states can surpass the inherent shot noise in an atom-based sensor and recent experiments have successfully created entangled Rydberg states in an OLC by exploiting the states' exaggerated principal quantum number[2, 3]. Hence, creating a system consisting of Rydberg entangled states in an OLC using strontium atoms could push the boundary of precision when using it for quantum metrology. Such a system could see wide applications such as in detecting dark matter candidates[4] or gravitational waves[5] and therefore I find the research fascinating. Although I have outlined one research interest, I am open to any projects in the domain of quantum metrology.

Relevant Experience

I have the relevant skills and knowledge necessary for tackling a PhD project in quantum sciences: programming, hands-on lab skills, and an aptitude for analysing topics in the domain.

I possess ample programming experience from previous projects within and outside my degree. In my LGAD work, I have created a Python script to calculate LGAD timing resolution based on measurements obtained from a dedicated experiment using Sr-90. Furthermore, I have designed a script used to perform automated measurements of an accelerator mass spectrometer (AMS) at HZDR using the same language. Aside from Python, I have experience using C++ to analyse test-beam data and contribute to Allpix², an open-source simulation framework for semiconductor sensors.

I have extensive hands-on lab experience. For example, since 2019, I have been working in the University of Glasgow's clean room where I have built a dedicated Sr-90 experiment for timing measurement of LGAD. In the same work, I have set up a laser and optical system to perform transient current technique measurements to characterise the gain of LGAD. Additionally, I have been involved in an engineering task of assembling a test beam-line prototype of an AMS. Experience in handling and building hardware for research has made me particularly adept at troubleshooting and improving experiments and will be highly beneficial for working with experiments in quantum sciences.

Lastly, I have experience handling and analysing topics in quantum mechanics. For instance, I have done my thesis on quantum theory exploring the Casimir-Polder force near granular metal films. Particularly, I had success analysing and digesting results from a variety of literature to derive the equation for the Casimir-Polder potential from first principle in the presence of a granular metal film. Then, numerical integration was done using experimental data available on the material's relative permittivity, and finally, I proposed an experiment to test the theory's predictions.

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References

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